Model Progress Report: K-Nearest Neighbors (KNN)

Task ID: TB-L09 **Date:** 05-Sep **Target:** Loan_Approval_Status COMPLETED

BASELINE (K=5)	OPTIMAL HYPERPARAMETER	PEAK TEST ACCURACY
77.50%	k = 8	81.50%
Default configuration	Grid evaluated k=1..20	Holdout test set

1. PIPELINE & PREPROCESSING IMPLEMENTATION

- **Target Isolation:** Separated `Loan_Approval_Status` as the binary target.
- **Feature Encoding:** Processed categorical columns (`Gender`, `Loan_Purpose`, `Loan_Type`) via one-hot encoding into **13 numerical dimensions**.
- **Dataset Partitioning:** Applied an 80/20 train-test split (**800 training rows**, **200 testing rows**) using `random_state=42` and `stratify=y`.
- **Feature Standardization:** Fitted `StandardScaler` on `X_train` and transformed `X_test` to guarantee scale-invariant Euclidean distance computation.

2. HYPERPARAMETER OPTIMIZATION & MODEL PERFORMANCE

K-Value Search: Evaluated k values from $k = 1$ to $k = 20$, plotting the K vs. Accuracy curve to observe bias-variance trade-offs. The peak test accuracy of **81.50%** was obtained at **$best\_k = 8$**.

Class	Precision	Recall	F1-Score
Class 0 (Not Approved)	0.82	0.97	0.89
Class 1 (Approved)	0.79	0.31	0.45
Overall / **Accuracy**		81.50%	

Confusion Matrix (N = 200)		
Actual \ Pred	**Pred: 0**	**Pred: 1**
Actual: 0	TN = 148	FP = 4
Actual: 1	FN = 33	TP = 15

3. KEY OPERATIONAL OBSERVATIONS

- **High Precision on Approvals (79%):** Out of all predicted approvals, 79% were correctly approved with only **4 False Positives**, significantly minimizing the institutional default risk.
- **Conservative Approval Recall (31.25%):** The model prioritizes conservative decision-making, resulting in **33 False Negatives** (eligible borrowers who were not approved).

4. SERIALIZED EXPORT ARTIFACTS

- 💾 `knn_model.pkl` (Trained `KNeighborsClassifier` instance, k=8)
- 💾 `scaler.pkl` (Fitted `StandardScaler` instance required for inference scaling)
- 💾 `model_columns.pkl` (Feature alignment column index with 13 dimensions)